

Selective Information Display Using High-Speed Projector and PDLCS

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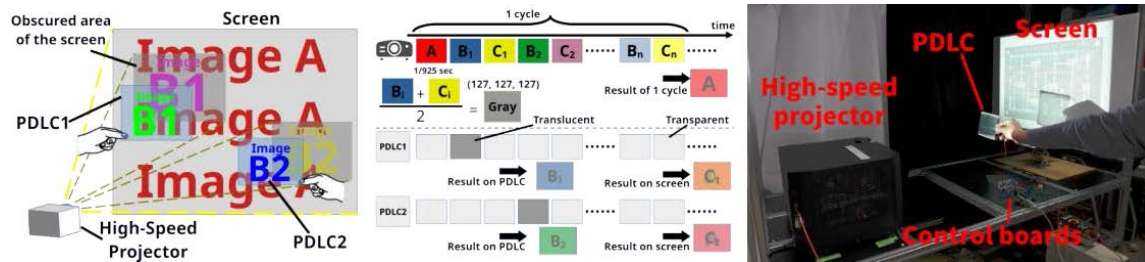


Figure 1: Left: System overview. By obscuring the projected image by PDLC, the projected image and obscuring frame on PDLC change. Center: Description of projected frames. Right: the picture of an example of projection.

ABSTRACT

This research proposes a novel technology that can change displayed images selectively by placing a PDLC on the screen. This system consists of a high-speed projector and some PDLCS. By intentionally obstructing the projected image, the different images appear on the surface of the PDLC and the screen. This technology is achieved by synchronizing the frequency of projected frames with the transparency of the PDLCS. The projected images contain the calculated inverted frames to display the desired image when no PDLC is held on the screen. This paper explains the method and implementation of a prototype and introduces two application examples: the information visible only through the device, and select arbitrary layer from multi-layered information.

Index Terms: Spatial augmented reality, high-speed projection, interaction device, PDLC.

1 INTRODUCTION

Information presentation using projectors is used in various contexts. Unlike flat screen displays, projector-based systems can make some real objects screen, such as walls and ceilings [3, 6]. Some methods exist for interacting with those projected information. One of the methods is using cameras to recognize hand gestures [1] or AR-markers fixed at projection surfaces [6]. In these methods, many ways to interact with real objects exist, such as touch interaction by hand tracking, placing or moving AR markers as interactions [9], using the real objects with markers [7], and grasping and moving hand-held projectors. However, these methods generally require tracking the user's hand position and markers using cameras, additional devices, and calibration. Additionally, shadows may appear on the screen due to the projection being obstructed by hands. Some research has been conducted to overcome occlusion issues or use the occlusion as a interaction method using multiple projectors [2], but the complexity of the system and the need for additional calibration can arise.

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This research proposes a system that enables interaction and overcoming these issues using PDLC, a film-shaped material that can change its transparency by adding voltage. By obstructing the projected image using PDLCS, the projected image on the PDLC surface and the screen can change. This system requires no calibration and allows for different images to be projected even on the obstructed portion. There are several studies using a PDLC as a screen. For instance, Sugiura et al. [8] proposed a wearable device that consists of a projector, a PDLC, and a retroreflective material. Their device can project two types of information that can be visible only from a specific direction and a wider range simultaneously. Li et al. [4] proposed the selective presentation of 2D and 3D images by using a polarization-dependent liquid crystal lens array and a PDLC. While their studies focused on presenting two types of information, the proposal of this paper aims to display three or more types of information using the difference of time-phase of PDLCS' transparency and to enable users to change parts of the projected image proactively. Nakamura et al. [5] proposed a method that uses one projector inside a cube-shaped PDLC screen to project onto both the screen and walls. Their method makes it possible to project different images on two different surfaces using one projector. Compared with their method, this paper enables users to observe a specific image selectively using PDLC.

2 PROPOSAL METHOD

This research proposes a novel technology that can display information dynamically using a high-speed projector and PDLC (Polymer Dispersed Liquid Crystal). The system employs high-speed projection at up to 925 Hz and a PDLC material that alters transparency based on applied voltage. The proposed system is illustrated in Fig. 1 left and center. When projecting multiple images at hundreds of frames per second, the human eye perceives an average color by blending individual image pixels. Therefore, by displaying multiple images and generating inverted images so that the averaged image becomes the target image and displaying them at high speed, it is possible to present the target image with multiple pieces of information. Although human vision cannot perceive each image displayed at high speed, the system can selectively alter the visible image by intentionally making some frames invisible. By intentionally obstructing some frames, the system can dynamically alter the visible images. (Fig. 1 center).

In this research, PDLC is used for the occlusion technique.

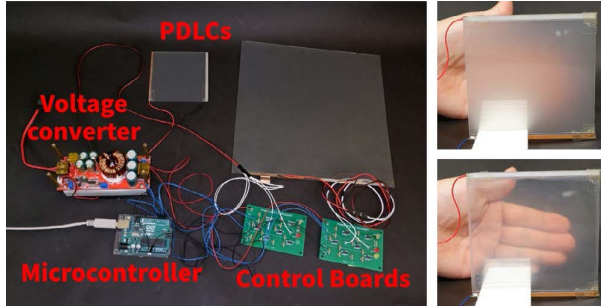


Figure 2: Left: The picture of implemented devices. Right: Pictures of PDLC. PDLC is a translucent material (upper). It turns transparent when applying a high voltage (lower).

PDLC is a translucent material that becomes transparent upon applying an 80V voltage and reverts to its translucent when the voltage is removed (Fig. 2). This material is positioned between the projector and the screen and switches its transparency synchronously with projected images of the high-speed projector to obstruct particular frames. PDLC is switched to translucent when the projector projects one specific frame. In another case, PDLC is transparent. Therefore, the image projected on the screen is changed to a different image which excludes one specific frame. In contrast, a particular frame appears on the surface of PDLC. Furthermore, it can display another different image on screen and PDLC by controlling the timing of transparent and translucent. Two or more PDLCs can display different images if the time phase of transparency differs. This method can change the image perceived by the user depending on the position and time phase of PDLC.

3 IMPLEMENTATION

3.1 Calculation of each frame

Consider a scenario where a projector displays image A on a screen and n of different time-shifted PDLC. Each PDLC panel displays a distinct image B_n . In this case, high-speed projector will project $2n + 1$ frames: $A, B_1, B_2, \dots, B_n$, and $C_1, C_2, \dots, C_n$, which cancel corresponding $B_1, \dots, B_n$ (Fig. 1 center). These $2n + 1$ frames are projected in one cycle. If one cycle is less than $1/60$ Hz of a second, each frame will be imperceptible to the human vision and perceived as the average of those. With the projection rate of 925 Hz, a maximum number of n is $n = 7$ because $925/60 = 15.4 \dots$. When the number of images n is relatively small, blank frames can be inserted to provide sufficient time for the PDLC to respond. Additionally, the canceling frames can potentially be consolidated, depending on the characteristics of A and $B_1, B_2, \dots, B_n$. Even in its transparent state, the PDLC has slight opacity. This leads to faint ghosting of the images intended to be visible on another PDLC or screen. This ghosting can be minimized by embedding the ghost image's canceling colors into other frames.

3.2 Hardware design

The prototype system consists of a high-speed projector (DynaFlash¹), computer, screen, PDLC (UMU FILM²), control boards for PDLC, and microcontroller. A system design is provided in Fig. 3 and the picture of devices is provided in the left of Fig. 2. This PDLC turns transparent in approximately 1 millisecond when applying the voltage, and returns to translucent in approximately 10 milliseconds when the voltage is removed. However, the response performance of PDLC depends on its shape and size, and electrical

¹<https://www.inrevium.com/product/tb-6v-dynaflash/>

²<https://umu.jp/product/film/index.html>

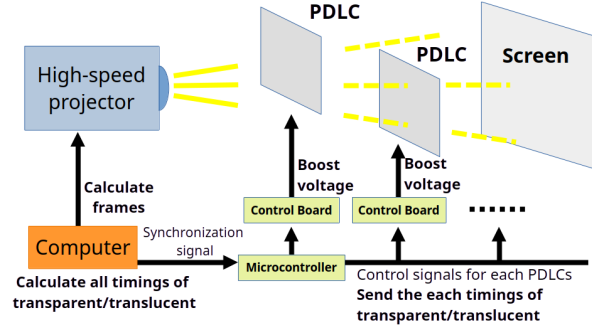


Figure 3: The overview of the implementation.

characteristics of the circuit. PDLC control board contains additional load resistors to enhance the response performance of PDLC. The computer generates multiple images to project and transmit to the high-speed projector. Simultaneously, a synchronization signal is sent to the microcontroller. The microcontroller sends the signal to each PDLC control board to switch transparent or translucent states of PDLC in proper timing depending on the synchronization signal. The PDLC control boards amplify the received signals to approximately 80 V. The example of projection setup in operation is provided in Fig. 1 right.

4 APPLICATIONS

4.1 The information visible only through the device

One potential application involves displaying information like maps. Fig. 4 shows the projection example of this application. A map without text is projected onto a screen. When a user holds up a PDLC-based device, corresponding place names appear only in the area covered by the device and on the PDLC itself. The user can select and obtain only the desired information on the map proactively. Furthermore, by using PDLCs with different time phases, it is possible to present different information to each PDLC. In this application example, the user can switch information in English and Japanese by using different PDLC-based devices. The high-speed projector projects five frames shown in Fig. 6 in this example.

4.2 Select arbitrary layer from multi-layered information

Multi-layered information exists in various contexts, such as printed circuit board designs or layered images in image editing software. By projecting images of each layer at a high frequency, all of the layers are blended and displayed on a screen, and users can selectively view a single layer of information by holding up a PDLC device. Fig. 5 shows an example of selecting and viewing a specific wiring layer from circuit board design data. In this example, four images B_1, B_2, B_3 , and C shown in Fig. 6 are projected sequentially. For such multi-layered information, the desired image $A \simeq B_1 + B_2 + B_3$. In this case, canceling frames C_1, C_2, C_3 can be reduced to one frame C .

5 DISCUSSION AND CONCLUSION

A primary limitation of this method is the constraint on the number of frames within one cycle that can be projected without being perceptible to humans. The high-speed projector used in this research can project images at 925 Hz. When projecting information at 60 Hz, only 15 frames can be embedded within a single cycle. Therefore, the current method can display a maximum of 7 different images. Furthermore, the images projected onto the PDLC are extracted from a specific frame within a cycle. As the number of frames within a cycle increases, the perceived brightness of the

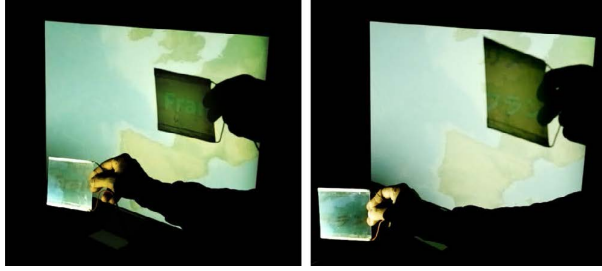


Figure 4: Example of map application. The information texts appear when PDLC obstructs the projection. Different information appears by using different PDLC

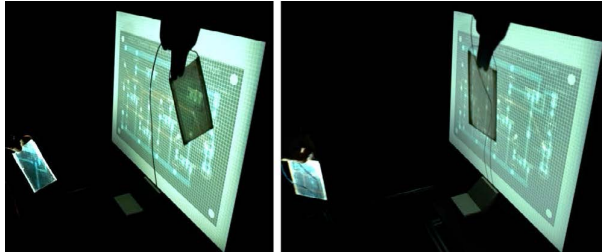


Figure 5: Example of multi-layered information application. The user can obtain a sufficient layer image using PDLC.

frame on PDLC decreases. The optical properties of the PDLC, such as reflectance and transmittance, must also be considered.

In this paper, we proposed and implemented a system that can change projected information selectively using a high-speed projector and PDLC. By synchronizing high-speed projection with the transparency of the PDLC, the system can change the displayed images. As the future works, we will explore further application possibilities and conduct user evaluations to assess the system's usability and effectiveness.

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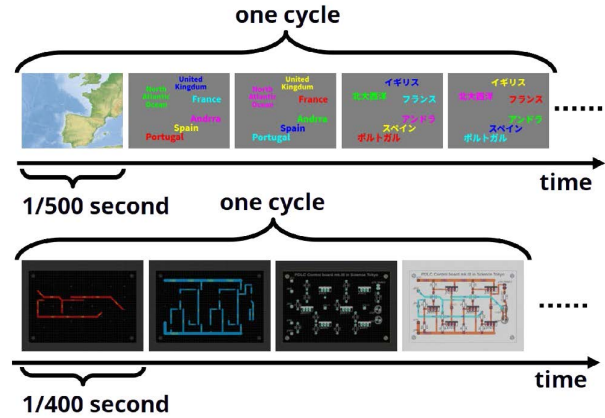


Figure 6: Projected images of two application examples.

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